



Yugesh Doddala

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Career Objective

Looking forward for the position in an organization where my technical and analytical skills are used for the betterment of the organization by enhancing my technical exposure.

Professional Career

Worked as **Team Lead** and **IP Verification** and **RTL Design Engineer** in **Young Minds Technology Solutions Pvt. Ltd**

Duration: April 2017 to August 2020

Experience Summary

Having Around 3+ Years of technical experience with expertise in **IP verification** and RTL design. During this period I worked with multiple domains like **Design Verification**, RTL and Analog Mixed Signal designing.

Roles & Responsibilities:

- ❖ Verification of IP blocks on SoC using system verilog and UVM methodology
- ❖ Executing various test cases to verifying functionality of the IP using system verilog and UVM
- ❖ Developing and executing assertion based verification plans for critical blocks
- ❖ Develop tests to achieve coverage goals and Outlined functional verification plans for ASIC Designs.
- ❖ Experience in RTL design and verification, Assisted RTL design engineers.
- ❖ Attended and participated in conferences and seminars as requested or required
- ❖ Served as a team lead for the VLSI during the tenure from June 2019 to August 2020
- ❖ Contributions towards Process Building, Strategic directions to Improve time to market
- ❖ Managed RTL design and verification, Assisted RTL design engineers.
- ❖ Plays a key role in improving work environment and recruiting new talent



Projects Handled

1. **UART IP Core Verification:**

HVL: System Verilog

TB Methodology: UVM

EDA Tool: Questasim Mentor Graphics

Description:

The UART IP core provides serial communication capabilities, which allow communication with modem or other external devices. UART will operate in 3 different modes-Simplex mode, Full duplex and loop back mode.

Responsibilities:

- ✓ Architected the class-based verification environment in UVM
- ✓ Defined Verification plan and Verified the RTL model using System Verilog
- ✓ Generated functional and code coverage for the RTL verification signoff

2. **Design and Verification of a Synchronous Dual Ported RAM in System Verilog**

HDL: Verilog

EDA Tool: Xilinx Vivado

Description:

This is A 64 bit x 8 bit synchronous, true dual-port RAM design with any combination of independent read or write operations in the same clock cycle in Verilog HDL.

Responsibilities:

- ✓ Architected block level structure of the design
- ✓ Implemented RTL using Verilog code and Synthesized the design
- ✓ Generated functional and code coverage for the RTL verification signoff

3. **Assertion based Verification of a Synchronous FIFO design**

HDVL: System Verilog

Tools Used: Questasim Mentor Graphics

Description:

A FIFO (First-In-First-Out) is a memory queue and has control logic with clock which efficiently manages read and write operations. It has the capability to notify the concerned modules regarding its empty status and full status to help ensure no underflow or overflow of data.

Responsibilities:



- ✓ Defined assertion based Verification plan
- ✓ Verified RTL model design using System Verilog
- ✓ Generated functional and code coverage for the RTL verification signoff

4. **Design and verification of an Master/Slave APB (Advanced Peripheral Bus)**

HDL: Verilog, HVL: System Verilog

Tools: Xilinx Vivado, QuestaSim, Cadence Encounter

Description:

The APB is from the AMBA protocol family which implements a low cost interface, minimizes the power consumption and reduces the interface complexity. The APB bus is designed using the verilog HDL according to the specification and is verified using Universal Verification Methodology.

Responsibilities:

- ✓ Architected the block level structure for the bridge.
- ✓ Developed Verilog RTL for each block.
- ✓ Verified each block with different transfers like single READ, WRITE & Burst READ, WRITE.
- ✓ Work with other engineers to debug and Synthesized the design

5. **Development of MAC unit with Adiabatic Logics**

Tools Used: Cadence Virtuoso, Mentor Tanner EDA

Technologies: TSMC 22nm, PTM 32nm

Description:

In this work, 32 bit MAC unit is implemented using ECRL Adiabatic logic for signal processing applications that promises a significant reduction in power consumption.

Responsibilities:

- ✓ Reviewed product requirements and schematics
- ✓ Work with other engineers to debug, characterize and optimize performance
- ✓ Perform new circuit R&D to be ahead of current technology

Education:

M.Tech (2016-2018) in Embedded Systems at Shree Rama Educational Group of institutions, Tirupati with 77%



B.Tech (2012-2016) in Electronics and Communications at Sree Rama Engineering college, Tirupati with 75.6% as aggregate.

Intermediate Education (2010-2012) at Narayana Junior College, Tirupati with 80%
Class X at P.C.R English medium High School, Tirupati with 72.6%

Competitive Skills

Languages Known: Telugu, English

Tool Expertise:

- ❖ Proven Experience with Industry standard tools like Cadence Virtuoso, Cadence Encounter, Mentor Questasim
- ❖ Having Experience with FPGA Tools specially like Xilinx ISE, Xilinx Vivado, and FPGA Boards like Xilinx Altera, Basys 3, Zed Board
- ❖ Some Other tools like Tanner EDA, MATLAB, OrCAD P-spice.

Programming Languages Known:

- ❖ Having Proven Experience with VHDL, Verilog HDL, System Verilog
- ❖ Python Beginner

Technical Competence:

- ❖ Strong Understanding of Digital Design and Verification
- ❖ Proven Experience in the areas of Synthesis and Timing
- ❖ Having Familiarity with Advanced verification methodologies like UVM and OVM
- ❖ Having experience with DFT, BIST, Scan test and JTAG
- ❖ Familiarity with ASIC and SoC Design Methodology
- ❖ Experienced with Advanced process nodes like 32nm, 45nm and 90nm
- ❖ Experienced with common Serial protocols like UART, SPI and AMBA, AXI
- ❖ Having Professional Experience as Trainer

**Personal Profile:**

Name: Doddala Yugesh
Date of Birth: 27-August-1994
Father's Name: D Balaji Naidu
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Declaration:

I Yugesh Doddala here by declare that all the information furnished above is true to the best of my knowledge and belief.

Place: Tirupati

(D. Yugesh)